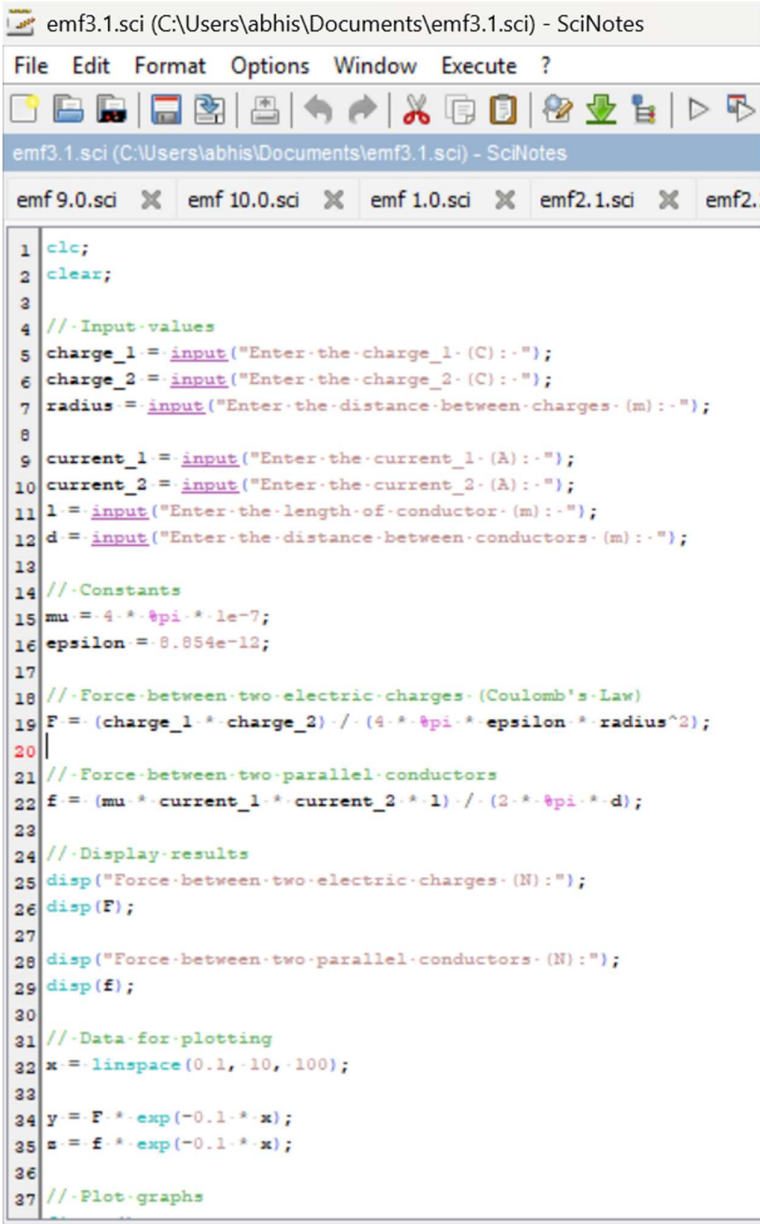


Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two

parallel conductors separated by varying distance.

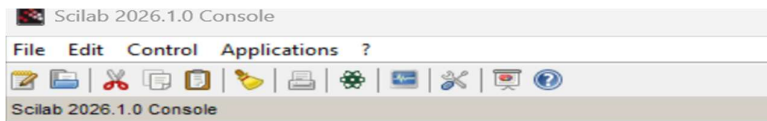
Aim: To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required: •SCILAB version 6.0.1



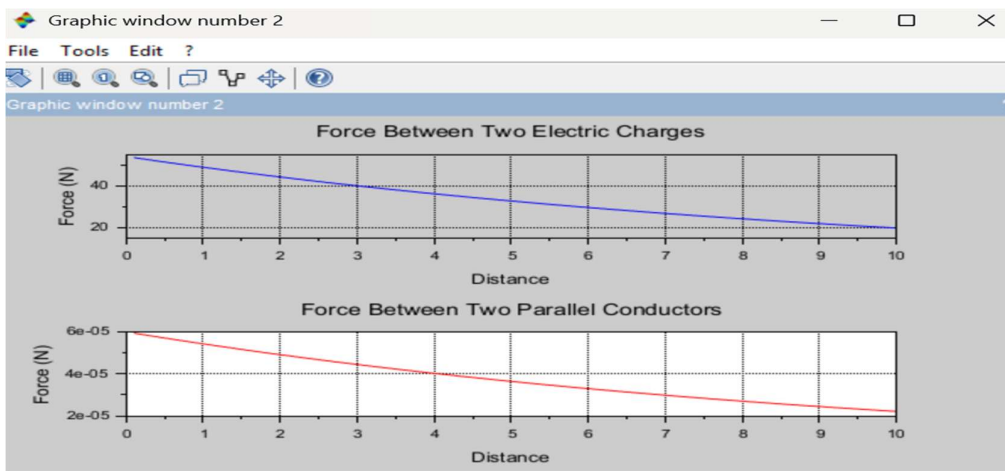
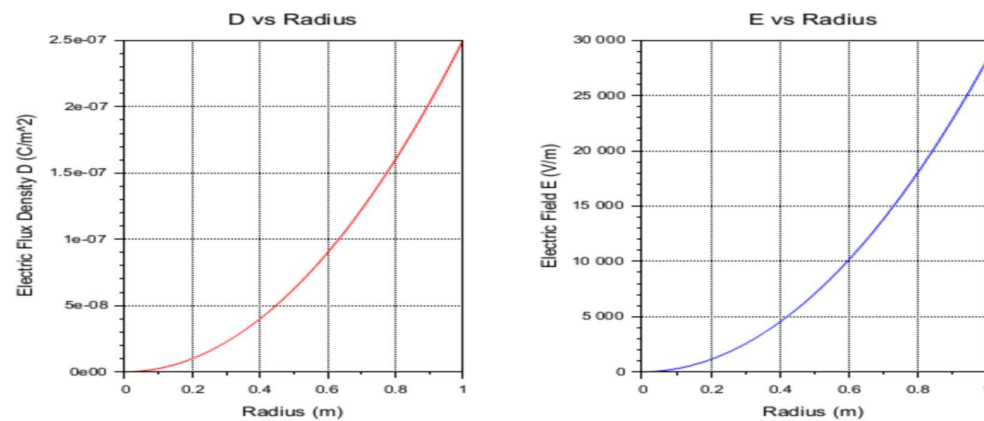
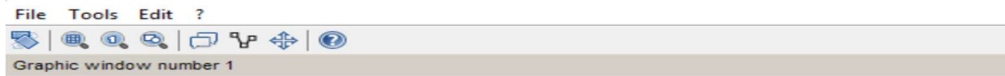
```
emf3.1.sci (C:\Users\abhis\Documents\emf3.1.sci) - SciNotes
File Edit Format Options Window Execute ?
emf3.1.sci (C:\Users\abhis\Documents\emf3.1.sci) - SciNotes
emf9.0.sci X emf10.0.sci X emf1.0.sci X emf2.1.sci X emf2.1.sci X

1 clc;
2 clear;
3
4 //Input-values
5 charge_1 = input("Enter the charge_1 (C) : ");
6 charge_2 = input("Enter the charge_2 (C) : ");
7 radius = input("Enter the distance between charges (m) : ");
8
9 current_1 = input("Enter the current_1 (A) : ");
10 current_2 = input("Enter the current_2 (A) : ");
11 l = input("Enter the length of conductor (m) : ");
12 d = input("Enter the distance between conductors (m) : ");
13
14 //Constants
15 mu = 4 * %pi * 1e-7;
16 epsilon = 8.854e-12;
17
18 //Force between two electric charges (Coulomb's Law)
19 F = (charge_1 * charge_2) / (4 * %pi * epsilon * radius^2);
20
21 //Force between two parallel conductors
22 f = (mu * current_1 * current_2 * l) / (2 * %pi * d);
23
24 //Display results
25 disp("Force between two electric charges (N) : ");
26 disp(F);
27
28 disp("Force between two parallel conductors (N) : ");
29 disp(f);
30
31 //Data for plotting
32 x = linspace(0.1, 10, 100);
33
34 y = F * exp(-0.1 * x);
35 z = f * exp(-0.1 * x);
36
37 //Plot graphs
```



```
Enter the charge_1 (C): 5*10^-6
Enter the charge_2 (C): 3*10^-6
Enter the distance between charges (m): 5*10^-2
Enter the current_1 (A): 5
Enter the current_2 (A): 4
Enter the length of conductor (m): 3*10^-2
Enter the distance between conductors (m): 2*10^-3

"Force between two electric charges (N):"
53.926455
"Force between two parallel conductors (N):"
0.00006
exec: Wrong number of output argument(s): 0 expected.
Enter the value of charge q (C):
```



③ Formula

$$F = \frac{q_1 q_2}{4 \pi \epsilon_0 r^2}$$

$$F = \frac{(2 \times 10^{-6})(4 \times 10^{-6})}{4 \pi (8.854 \times 10^{-12})(0.5)^2}$$

$$F = \frac{8 \times 10^{-12}}{2.781 \times 10^{-11}}$$

$$F = 0.287655$$

Formula $F = \frac{\mu_0 I_1 I_2}{2 \pi d}$

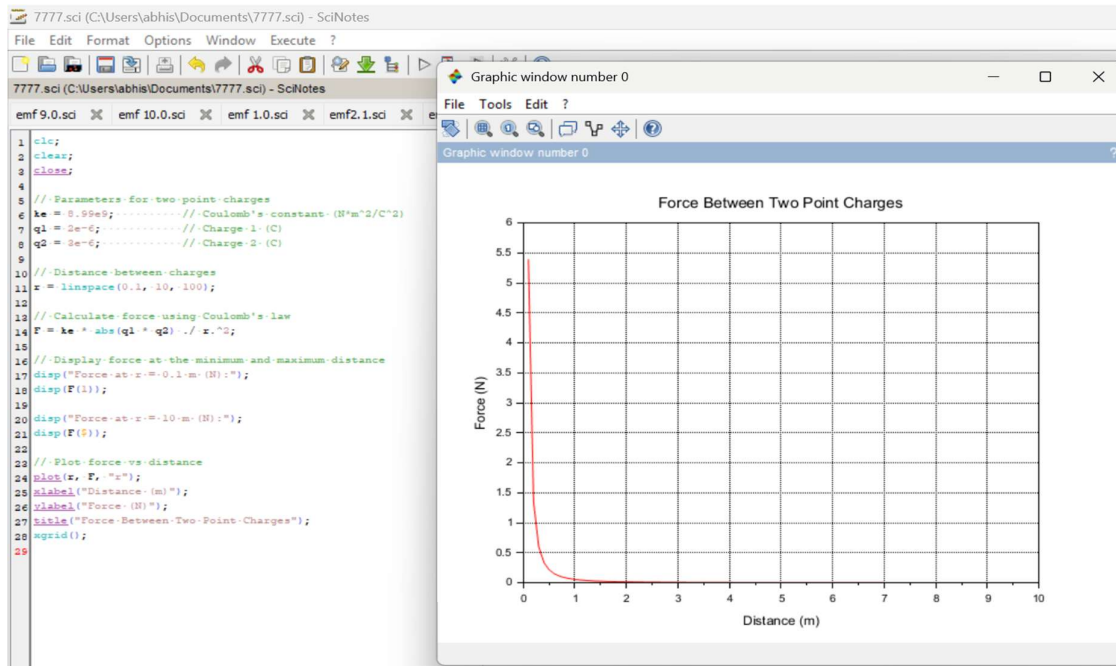
$$F = \frac{(4 \pi \times 10^{-7})(5)(8)(2)}{2 \pi (0.1)}$$

$$F = \frac{3.2 \times 10^{-5}}{0.2}$$

$F = 1.6 \times 10^{-4} \text{ N}$

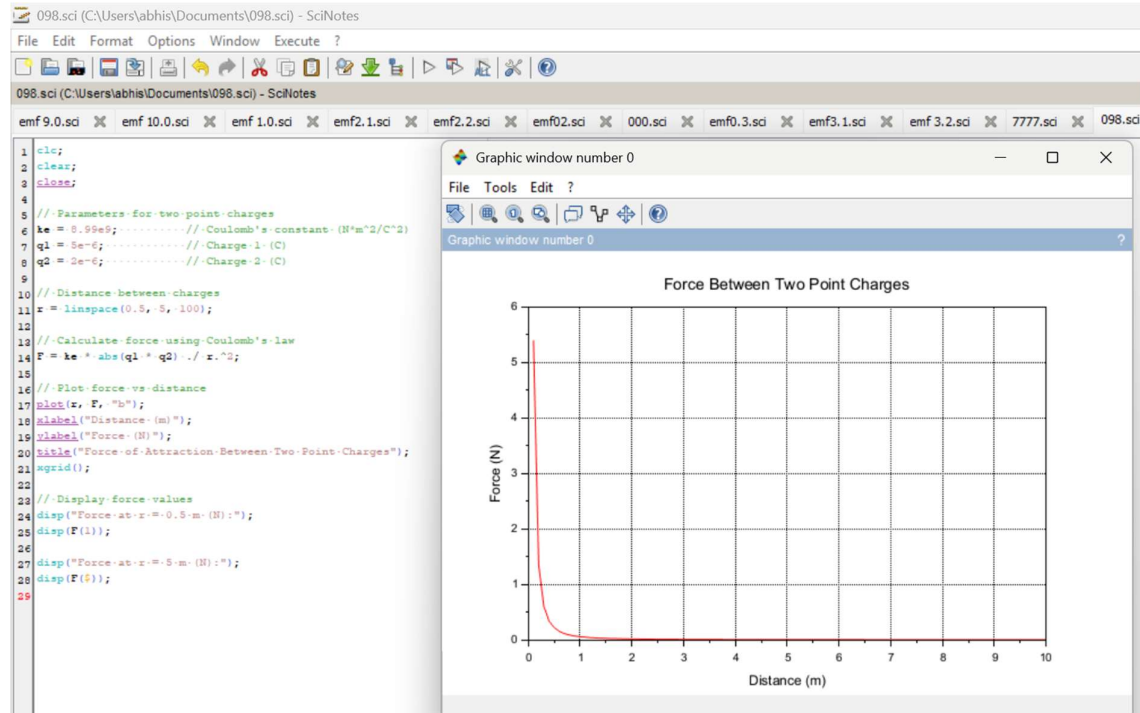
$F = 0.00006 \text{ N}$

Case 1 : Given two point charges $q_1 = 2 \times 10^{-6}$ and $q_2 = 3 \times 10^{-6} \text{ C}$, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.

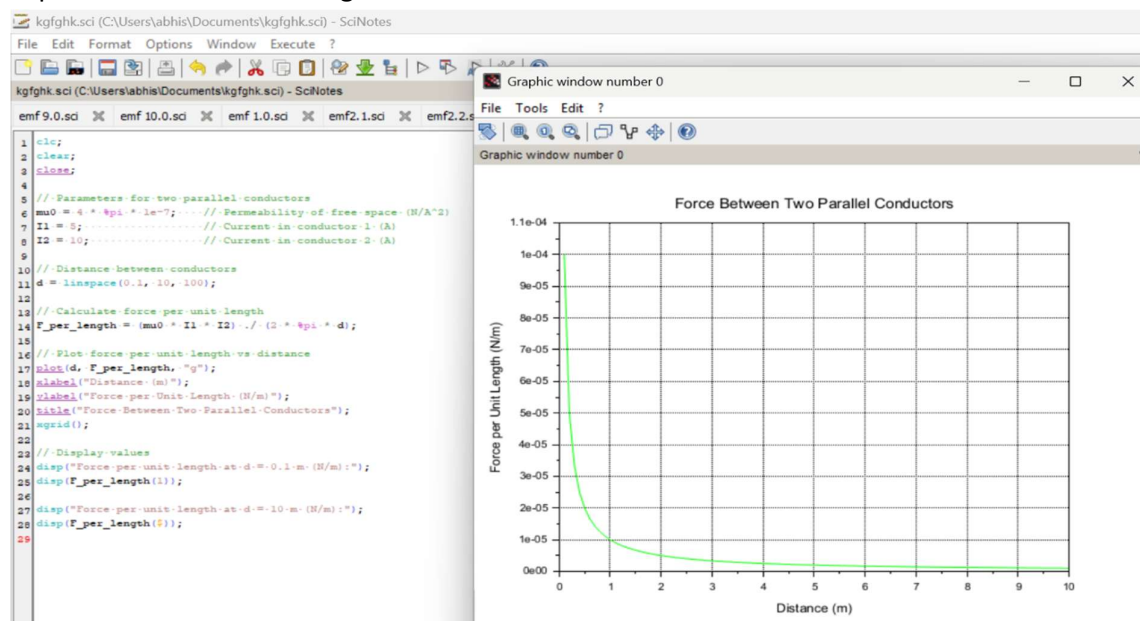


Case 2:

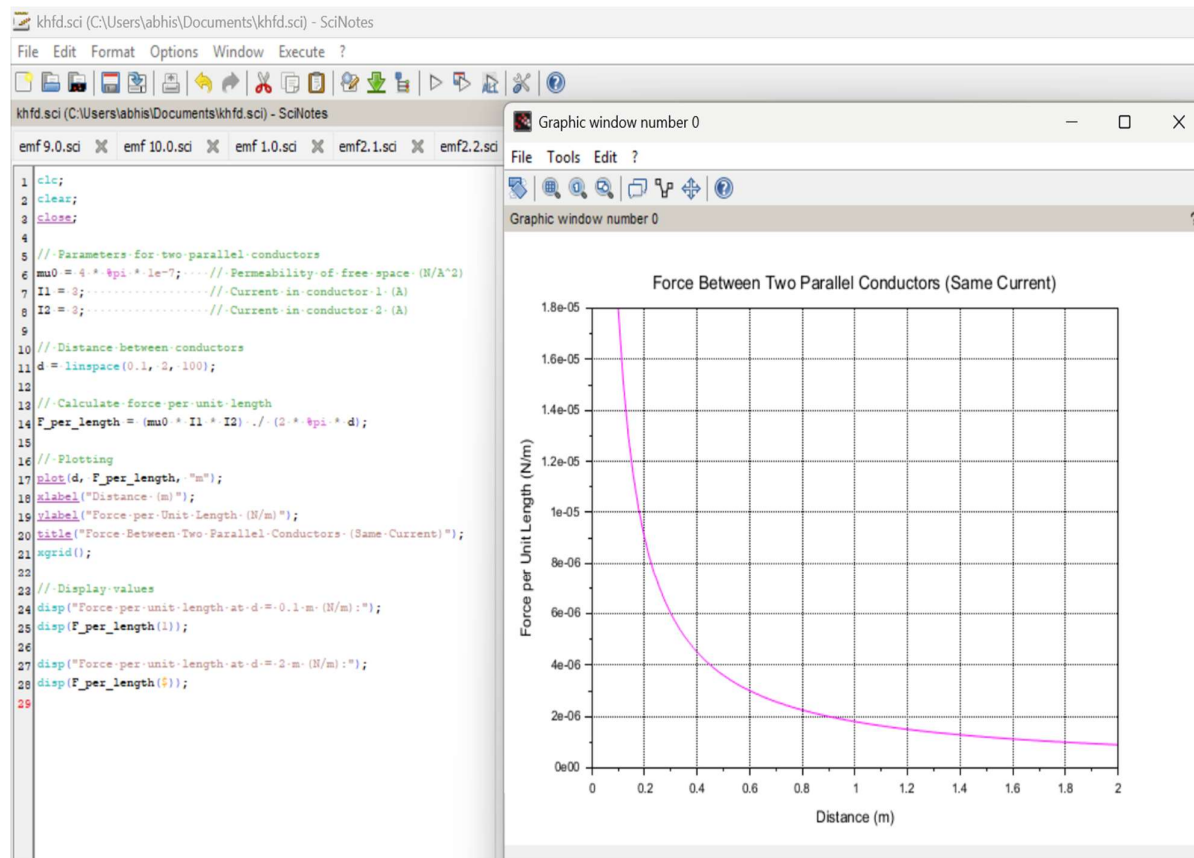
Given two point charges $q_1 = 5 \times 10^{-6} \text{ C}$ and $q_2 = 2 \times 10^{-6} \text{ C}$, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.



Case 3: Two parallel conductors are separated by a distance d . The first conductor carries a current of $I_1 = 5 \text{ A}$, and the second conductor carries a current of $I_2 = 10 \text{ A}$. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors.



Case 4: Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.



Result :Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.